Heterogeneous Households and Aggregate Dynamics: Analytical Moment Recursion[†]

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Abstract

I develop an analytical framework for heterogeneous-household economies with idiosyncratic labor productivity and wealth returns. I prove the existence of a moment-recursive competitive equilibrium (MRCE) in which aggregate capital and two distributional covariances follow a closed-form matrix recursion. The equilibrium holds regardless of household beliefs about aggregate dynamics, naturally accommodating non-stationary growth paths. This closed-form structure enables exact general equilibrium inversion: given U.S. data on output, capital, and hours (1948–2024), I recover structural shocks period by period without simulation or approximation. The inversion reveals state-dependent amplification from distributional covariances, a sorting accelerator through which TFP growth drives rising inequality, and a risk premium from idiosyncratic return heterogeneity.

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1 Introduction

The distribution of household income, wealth, and portfolio returns shapes aggregate output, employment, and asset prices (Kaplan et al., 2018). Yet the Bewley–Huggett–Aiyagari framework requires tracking the entire wealth distribution—an infinite-dimensional state variable. Krusell and Smith (1998) showed numerically that mean wealth nearly suffices for forecasting prices, but this approximation lacks an analytical foundation and breaks down with richer heterogeneity. This paper closes the gap. I construct a model in which three summary statistics—aggregate capital and two wealth-characteristic covariances—exactly characterize aggregate dynamics via a closed-form matrix recursion. I call the resulting object a *moment-recursive competitive equilibrium* (MRCE).

I make three contributions. First, I derive closed-form policies for consumption, saving, and labor supply under incomplete markets with idiosyncratic labor-productivity and return risk. These policies yield a unique MRCE with law of motion $\mathcal{M}' = \beta\,\widetilde{\mathcal{G}}\,\mathcal{M} + [T, 0, 0]'$, where $\mathcal{M} = [K, \operatorname{cov}(z, a), \operatorname{cov}(e, a)]$ and $\widetilde{\mathcal{G}}$ depends on current prices. The saving rate is increasing and concave in wealth—consistent with Dynan et al. (2004)—and the borrowing constraint is never binding. Shutting down return heterogeneity nests Krusell–Smith; shutting down all heterogeneity recovers Brock and Mirman (1972). Second, I show the equilibrium is invariant to household beliefs about aggregate dynamics: under log utility, income and substitution effects of future uncertainty cancel exactly, so policies depend only on current prices. This eliminates the fixed-point problem over beliefs, making the MRCE a belief-invariant recursive competitive equilibrium (BRCE). Because policies depend only on current prices, the equilibrium accommodates non-stationary aggregate paths without stationarity or detrending assumptions. Third, the closed-form structure enables an *exact general equilibrium inversion*. Given U.S. data on output, capital, and hours (1948–2024), I recover sequences of TFP, time-varying patience, and wealth-leisure complementarity that exactly rationalize the data period by period, without simulation or approximation. This extends the business-cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting with structural rather than reduced-form shocks.

The inversion yields several substantive findings. TFP captures the trend but explains

only 15 percent of cyclical volatility. The recovered discount factor is strongly countercyclical ($\text{corr}(\beta_t, \log Y_t) = -0.72$) and acts as a powerful stabilizer: removing it more than doubles output volatility. Distributional covariances generate state-dependent amplification: a 2 percent TFP shock produces a 37 percent larger output response from the high-inequality state of 2022 than from that of 1952. The mechanism is that wealth concentration intensifies the feedback between returns and saving. TFP growth acts as a "sorting accelerator": it raises wages and returns disproportionately for productive-wealthy households, generating the concentration patterns documented by Piketty (2014). The model also delivers a novel asset-pricing result: idiosyncratic return heterogeneity, traditionally viewed as diversifiable, commands a positive risk premium because wealth sorting alters aggregate labor supply and returns (Constantinides and Duffie, 1996; Storesletten et al., 2004).

Related literature. Bewley (1986), Huggett (1993), and Aiyagari (1994) developed the foundational heterogeneous-agent framework. Krusell and Smith (1998) introduced aggregate risk and showed numerically that mean wealth nearly suffices for forecasting prices. The MRCE provides an exact analytical counterpart—a finite-dimensional moment recursion—that accommodates return heterogeneity and clarifies when and why approximate aggregation succeeds or fails. Guvenen (2009) studies heterogeneous earnings dynamics; Kaplan et al. (2018) and Auclert et al. (2021) develop heterogeneous-agent New Keynesian models. The MRCE differs by delivering exact aggregation rather than relying on numerical solution or linearization.

The analytical structure builds on the tradition of closed-form aggregate dynamics. Brock and Mirman (1972) solve the stochastic growth model in closed form under log utility; the MRCE nests their solution. Benhabib and Farmer (1994) derive determinacy conditions in growth models; the MRCE establishes uniqueness of the moment recursion. Angeletos (2007) studies uninsured idiosyncratic investment risk—the return heterogeneity central here—but does not obtain a closed-form law of motion for the distribution. Bilbiie (2020) and Ravn and Sterk (2021) develop tractable heterogeneous-agent models with analytical aggregate dynamics; this paper uses a continuum of agents and delivers exact aggregation with rich return heterogeneity. Alvarez and Lippi (2022) develop analytical impulse responses using

eigenvalue methods in menu-cost models; the moment recursion plays an analogous role for heterogeneous-household economies.

The belief-invariance result relates to the literature on equilibrium refinement and bounded rationality. Menzio and Shi (2010) introduce block-recursive equilibria in which agents need not track the aggregate distribution; the BRCE goes further by showing invariance to beliefs about aggregate dynamics themselves. This connects to Angeletos and La'O (2013) on sentiments and Farhi and Werning (2019) on bounded rationality.

The state-dependent amplification results contribute to the literature on nonlinear business cycle dynamics. Storesletten et al. (2004) document countercyclical cross-sectional risk, Bloom et al. (2018) study how uncertainty generates state-dependent aggregate responses, and Berger and Vavra (2015) analyze the role of wealth heterogeneity for consumption dynamics. The MRCE analytically characterizes how distributional covariances amplify or dampen aggregate shocks.

Finally, the asset pricing results relate to the literature on idiosyncratic risk and equilibrium returns (Constantinides and Duffie, 1996; Schmidt, 2016). Growing evidence on heterogeneous portfolio returns (Fagereng et al., 2020; Bach et al., 2020) motivates the model's return channel, which generates an aggregate risk premium through wealth sorting.

Outline. Section 2 presents the model. Section 3 derives the MRCE, establishes belief invariance, and analyzes macroeconomic and asset pricing implications. Section 4 presents the quantitative analysis. Section 5 discusses extensions. Section 6 concludes. All proofs and supplementary results appear in the Online Appendix.

2 Model

The economy is populated by a continuum of infinitely lived households who face idiosyncratic labor productivity and portfolio return risk and choose consumption, saving, and labor supply each period.

2.1 States and shocks

Each household's individual state is $x = [a, z, e]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous), and an idiosyncratic factor e that scales the household's return on capital (exogenous). All three dimensions display substantial cross-sectional variation in the data (Kuhn et al., 2020; Storesletten et al., 2004; Fagereng et al., 2020).

The aggregate state is $X = [\Phi, A, B, C]$, where Φ is the cross-sectional joint distribution of individual states, A is aggregate TFP, B governs the dispersion of capital returns, and C governs the dispersion of labor productivity. All three exogenous processes follow Markov chains. In the quantitative application (Section 4), only A varies over time; B and C are held at steady state. The framework nevertheless accommodates time-variation in all three, allowing the structure of inequality itself to shift—for instance, a widening of the return distribution during financial crises or a compression of the wage distribution following skill-biased technological change.

2.2 Preferences

Household preferences take a Stone-Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \tag{1}$$

where c is consumption, n is labor supply, and $\theta > 0$ governs the relative weight on leisure. Two features merit discussion.

First, the subsistence consumption level $g(x; X) = \kappa w(X)(1-\tau)z$ is proportional to after-tax potential earnings, capturing work-related expenses—commuting, child care, professional attire—that scale with take-home pay.

Second, the effective time endowment $\kappa + \xi a$ increases in wealth: $\kappa \in (0, 1)$ is the base endowment, and $\xi > 0$ captures the degree to which wealth buys usable time through time-saving services or home production (Aguiar and Hurst, 2007). This wealth-leisure complementarity is the primary channel through which the wealth distribution affects aggregate labor supply.

The three parameters (θ, κ, ξ) are calibrated to match average hours, the saving-to-income

ratio, and the wealth Gini. Extensions to GHH and Epstein-Zin preferences appear in Section 5.

2.3 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x;X) = \max_{c,a',n} \quad u(c,n,a,z;X) + \beta \mathbb{E} v(x';X') \tag{2}$$

$$\text{s.t.} \quad c + a' = w(X)(1-\tau)zn + (1 + r(X)(1-\tau)e)a + T(X) \tag{3}$$

$$\Phi' = \Gamma_\Phi(X) \tag{4}$$

$$a' \geq 0, \quad n \geq 0 \tag{5}$$

Consumption plus saving equals after-tax labor income $w(X)(1-\tau)zn$, after-tax capital income $(1+r(X)(1-\tau)e)a$, and a lump-sum transfer $T(X)$. The wage $w(X)$ is per efficiency unit of labor and $r(X)$ is the aggregate return on capital. The household-specific factor e scales this return: $e > 1$ means an above-average return, $e < 1$ a below-average one, and $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average return.

The household discounts the future at rate $\beta \in (0,1)$, faces a borrowing constraint $a' \geq 0$, and takes the law of motion Γ_Φ for the aggregate distribution as given. As shown below, this constraint is non-binding in equilibrium—a consequence of the additive transfer $T(X)$ and the multiplicative structure of the saving policy.

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = AK^\alpha L^{1-\alpha}, \tag{6}$$

where $K = \int a \, d\Phi$ is the aggregate capital stock, $L = \int zn(x;X) \, d\Phi$ is aggregate effective labor, and $\alpha \in (0,1)$ is the capital share. Factor prices are determined by the first-order

conditions of the firm's problem:

$$r(X) = \alpha A K^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1-\alpha) A K^{\alpha} L^{-\alpha}, \tag{7}$$

where δ is the depreciation rate. Return heterogeneity redistributes income across households without affecting total output. Because factor prices depend on K/L, any distributional force that shifts labor supply feeds back into prices.

2.5 Government policy

The government levies a proportional income tax at rate τ on both capital and labor income, and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot Y(X) \cdot \tau, \tag{8}$$

where $Y(X) = AK^{\alpha}L^{1-\alpha}$ is aggregate output. The remaining revenue $(1-\eta) \cdot Y \cdot \tau$ finances government consumption, which does not enter household utility or the production function.

The additive transfer T plays a critical theoretical role: it provides the intercept in the Kesten process governing individual wealth dynamics, ensuring a well-defined stationary wealth distribution with a Pareto tail (Kesten, 1973). The parameter η controls redistribution: a higher η compresses the distribution, while a lower η allows greater dispersion and a thicker tail.

2.6 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_Φ.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*

4. *Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ.*

The difficulty is that Φ is infinite-dimensional. The next section shows that optimal household policies depend on Φ only through a three-dimensional vector of moments, and that these moments follow an explicit matrix recursion—collapsing the infinite-dimensional state to a finite system.

3 Moment-Recursive Competitive Equilibrium

This section derives the main theoretical results: closed-form household policies, the existence and uniqueness of the MRCE, belief invariance, the macroeconomic decomposition, and asset pricing implications.

3.1 Closed-form household policies

The household's optimal consumption, saving, and labor supply all admit closed-form expressions. These analytical policies are the foundation of all subsequent results: they allow the infinite-dimensional distribution to be replaced by a finite vector of moments and make the equilibrium law of motion an explicit matrix recursion rather than an implicitly defined fixed point.

Proposition 1 (Analytical household-level decisions)**.**
The optimal consumption, saving, and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1 + r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)z \tag{9}$$

$$a'(x;X) = \beta a\left((1 + r(X)(1-\tau)e) + w(X)(1-\tau)z\xi\right) + T(X) \tag{10}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)z},\ 0\right\} \tag{11}$$

Proof. The budget constraint can be rewritten as:

$$(1+\theta)(c-\kappa z w(X)(1-\tau))+a' = ((1+r(X)(1-\tau)e)+w(X)(1-\tau)z\xi)\,a+T(X). \quad (12)$$

The transfer $T(X)$ is identical across households and therefore enters savings as an additive intercept. The result of Brock and Mirman (1972) immediately applies to the multiplicative component. ■

The consumption function is linear in wealth, but the marginal propensity to consume differs across income sources—higher out of capital income than out of labor income. The saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component, which enables analytical aggregation, and an additive transfer, which ensures a stationary wealth distribution.

3.2 The saving rate

The saving rate can be characterized in closed form, connecting the model directly to empirical evidence on saving behavior.

Proposition 2 (Analytical saving rate)**.**
The saving rate $s(x;X) = \frac{a'(x;X)-a}{y(x;X)}$*, where* $y(x;X) = w(X)(1-\tau)zn(x;X)+r(X)(1-\tau)ea+T(X)$ *is after-tax income plus transfer, satisfies:*

$$s(x;X) = \frac{a[\beta(1+r(X)(1-\tau)e+w(X)(1-\tau)z\xi)-1]+T(X)}{w(X)(1-\tau)z\kappa\theta + a\left[\frac{w(X)(1-\tau)z\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+r(X)(1-\tau)e)}{1+\theta} + r(X)(1-\tau)e\right]+T(X)}, \quad (13)$$

with the following properties:

(i) $\lim_{a\to\infty} s(x;X) = \frac{\beta(1+r(X)(1-\tau)e+w(X)(1-\tau)z\xi)-1}{\frac{w(X)(1-\tau)z\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+r(X)(1-\tau)e)}{1+\theta} + r(X)(1-\tau)e}$

(ii) $\lim_{a\to 0} s(x;X) > 0$ *due to the transfer* $T(X) > 0$

(iii) $\frac{\partial s}{\partial a} > 0$ *and* $\frac{\partial^2 s}{\partial a^2} < 0$.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive even at zero wealth (due to the transfer), increases in wealth, and is concave. Richer households save a larger fraction of their income, but the marginal increase in the saving rate diminishes—precisely the pattern documented by Dynan et al. (2004) in the Consumer Expenditure Survey and Survey of Consumer Finances.

The concave saving rate generates mean reversion in relative wealth positions: the subsistence floor absorbs a diminishing share of total resources as wealth increases, so the saving rate converges to a constant in the limit.

3.3 Non-binding borrowing constraint

An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint)**.**
The borrowing limit $a' \geq 0$ is always non-binding.

Proof. Given $a'(x; X) = \beta a((1 + r(X)(1 - \tau)e) + w(X)(1 - \tau)z\xi) + T(X) \geq T(X) > 0$, the result is immediate. ■

The non-binding constraint is critical for tractability: it makes policy functions globally smooth and linear, enabling analytical aggregation.

3.4 Existence and characterization

I now define the central equilibrium concept.

Definition 2 (Moment recursive competitive equilibrium (MRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ is a moment recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_M.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*

4. *Γ_M satisfies $\Gamma_M(\mathcal{M}(\Phi), A) = \mathcal{M}(\Phi')$, where $\mathcal{M}$ extracts the sufficient moments from Φ, and Φ' is the distribution of individual states induced by g_a and the exogenous processes.*

The key departure from the RCE is that Γ_M maps a *finite* moment vector forward, rather than the entire distribution.

Proposition 3 (Unique existence of the MRCE)**.**
The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z,a), cov(e,a)]$, and the law of motion Γ_M is as follows. Define after-tax prices $\tilde{r} = r(\mathcal{M}, A)(1-\tau)$ and $\tilde{w} = w(\mathcal{M}, A)(1-\tau)$, and the transfer $T = \eta \cdot Y \cdot \tau$. Then:

$$\Gamma_M(\mathcal{M}(\Phi), A) = \begin{bmatrix} K' \\ cov(z', a') \\ cov(e', a') \end{bmatrix} = \beta \widetilde{\mathcal{G}} \begin{bmatrix} K \\ cov(z, a) \\ cov(e, a) \end{bmatrix} + \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix} \tag{14}$$

where the transition matrix is

$$\widetilde{\mathcal{G}} = \begin{bmatrix} 1 + \tilde{r} + \tilde{w}\xi & \tilde{w}\xi & \tilde{r} \\ \phi_z \rho_z \tilde{w} \xi \sigma_z^2 & \phi_z \rho_z (1 + \tilde{r} + \tilde{w}\xi) & 0 \\ \phi_e \rho_e \tilde{r} \sigma_e^2 & 0 & \phi_e \rho_e (1 + \tilde{r} + \tilde{w}\xi) \end{bmatrix} \tag{15}$$

where prices satisfy:

$$r(\mathcal{M}, A) = \alpha A \left(\frac{L}{K}\right)^{1-\alpha} - \delta \tag{16}$$

$$w(\mathcal{M}, A) = (1-\alpha) A \left(\frac{K}{L}\right)^{\alpha} \tag{17}$$

$$L(\mathcal{M}, A) = \kappa + \frac{\xi(1+\theta\beta)(K + cov(z,a))}{1+\theta} - \frac{\theta(1-\beta)}{(1+\theta)\tilde{w}}[K(1+\tilde{r}) + cov(e,a)\tilde{r}] \tag{18}$$

The transfer T is identical across households and therefore enters only the K' equation. It drops out of the covariance equations because $cov(z', T) = cov(e', T) = 0$.

Proof. See Appendix A.3 of the Online Appendix. ■

This proposition is the main result of the paper. The state vector $\mathcal{M} = [K, \mathrm{cov}(z,a), \mathrm{cov}(e,a)]$ comprises aggregate capital and two distributional covariances that measure the alignment

between wealth and idiosyncratic characteristics. Its law of motion is a non-homogeneous linear recursion in which the off-diagonal entries of $\widetilde{\mathcal{G}}$ capture cross-moment feedback.

Two features of $\widetilde{\mathcal{G}}$ deserve emphasis. First, the $(1,3)$ entry $\tilde{r}$ shows that return-wealth sorting raises future capital, while the $(3,1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that higher capital strengthens return-wealth sorting—a self-reinforcing loop. Second, the $(2,3)$ and $(3,2)$ entries are zero because z and e are independent: the two sorting mechanisms interact only through aggregate capital. The signal-to-noise ratios ϕ_z and ϕ_e govern the speed of distributional adjustment. A detailed interpretation appears in Appendix B.1–B.2.

3.5 Ergodicity

Corollary 2 (Recursivity of MRCE and Kesten stationarity)**.**
The law of motion $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is a non-homogeneous multivariate Kesten process with both multiplicative ($\beta\widetilde{\mathcal{G}}$) and additive ($T$) components. When the spectral radius $\rho(\beta\widetilde{\mathcal{G}}) < 1$, the system admits a unique stationary solution

$$\mathcal{M}^* = (I - \beta\widetilde{\mathcal{G}})^{-1} \begin{bmatrix} T \\ 0 \\ 0 \end{bmatrix}. \tag{19}$$

The additive transfer intercept $T > 0$ is essential: without it, the purely multiplicative system $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M}$ has no well-defined stationary distribution, as the Kesten process requires both multiplicative and additive components to generate ergodic dynamics with Pareto tails in the wealth distribution (Kesten, 1973).

Proof. See Appendix A.4 of the Online Appendix. ■

The economy therefore converges to a unique stationary state $\mathcal{M}^*$, which serves as the starting point for business cycle analysis. The speed of convergence is governed by the spectral radius of $\beta\widetilde{\mathcal{G}}$. As shown in Section 4, near-unit-root behavior in this spectral radius generates slow mean reversion and amplifies the propagation of aggregate shocks.

3.6 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue)**.**
Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a)]$, and the law of motion is:

$$\Gamma_M(\mathcal{M}(\Phi), A) = \begin{bmatrix} K' \\ cov(z', a') \end{bmatrix} = \begin{bmatrix} \beta\left[(1 + \tilde{r} + \tilde{w}\xi)K + \tilde{w}\xi\, cov(z, a)\right] + T \\ \phi_z \rho_z \beta\left[(1 + \tilde{r} + \tilde{w}\xi)\, cov(z, a) + \tilde{w}\xi\sigma_z^2 K\right] \end{bmatrix} \tag{20}$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \mathrm{cov}(z, a)]$—the analytical counterpart of the Krusell and Smith (1998) economy. Aggregate capital alone is sufficient if and only if $\mathrm{cov}(z, a) = 0$; the magnitude of this covariance determines when approximate aggregation succeeds and when it fails. Further discussion appears in Appendix B.3.

Corollary 4 (Representative-agent analogue)**.**
Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w}\xi)K\right] + T, \tag{21}$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A) = \frac{1 - \beta}{1 + \theta}(1 + \tilde{r} + \tilde{w}\xi)K + \kappa\tilde{w}. \tag{22}$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mirman, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

3.7 Belief invariance

A distinctive feature of the model is that the equilibrium is invariant to household beliefs about aggregate dynamics, eliminating the fixed-point problem over beliefs that dominates computation in standard heterogeneous-agent models.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE)).
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ *is a belief-invariant recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.*
2. K, L *satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. Γ_Φ *satisfies* $\Gamma_\Phi(X) = \Phi'$, *where* Φ' *is the future distribution of individual states constructed by applying the policy function* g_a *and the exogenous idiosyncratic process to* Φ.

Theorem 1 (Belief invariance).
In the BRCE, the law of motion Γ_Φ *is unaffected by households' expectations about future aggregate states.*

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x; X) = \beta a((1+r(X)(1-\tau)e)+w(X)(1-\tau)z\xi)+T(X)$ depends only on current prices and the household's own state—not on forecasts of future prices. Under log utility, the income and substitution effects of anticipated changes in future returns exactly offset, so each household's policy is belief-invariant. Because all individual policies share this property, the aggregate law of motion inherits it.

Proposition 4 (Unique existence of the BRCE).
The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

Because the MRCE and the BRCE coincide, growth and business cycle dynamics can be studied without specifying how households form expectations. The equilibrium is computed by forward recursion from any initial state $\mathcal{M}_0$ and shock sequence $\{A_t\}$, with no fixed-point iteration required. This property enables the exact general equilibrium inversion of Section 4: structural shocks can be recovered from data without taking a stand on expectation formation.

3.8 Macroeconomic implications

I now analyze how the cross-sectional distribution shapes aggregate outcomes. The analytical structure yields an exact decomposition of aggregate dynamics into a representative-agent component and a distributional component.

State-dependent aggregate dynamics. The law of motion for aggregate capital is:

$$K' = \beta\left[(1+\tilde{r}+\tilde{w}\xi)K + \tilde{r}\,\mathrm{cov}(e,a) + \tilde{w}\xi\,\mathrm{cov}(z,a)\right] + T \tag{23}$$

The first term, $\beta(1+\tilde{r}+\tilde{w}\xi)K$, is the representative-agent channel—the capital accumulation that would obtain if all households were identical. The second term, $\beta\tilde{r}\,\mathrm{cov}(e,a)$, is the *return-sorting channel*: because high-return households are also disproportionately wealthy, their saving generates capital accumulation beyond what average returns and average wealth would predict. The third term, $\beta\tilde{w}\xi\,\mathrm{cov}(z,a)$, is the *productivity-sorting channel*: when productive households hold more wealth, their wealth-leisure complementarity generates additional labor income that is partially saved. These distributional terms make aggregate dynamics *state-dependent*: the same productivity shock produces different investment, consumption, and employment responses depending on the current degree of wealth sorting. When both covariances are large, a larger share of saving is concentrated among wealthy households whose behavior is more sensitive to returns, so aggregate investment responds more strongly to changes in prices.

Distributional channels. The aggregate consumption function decomposes as:

$$C(\mathcal{M}, A) = \underbrace{\frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}\xi)K + \kappa\tilde{w}}_{\text{Representative-agent level}} + \underbrace{\frac{1-\beta}{1+\theta}\left(\tilde{r}\,\mathrm{cov}(e,a) + \tilde{w}\xi\,\mathrm{cov}(z,a)\right)}_{\text{Distributional channel}} \tag{24}$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M}, A) = \kappa + \underbrace{\frac{\xi(1+\theta\beta)(K+\mathrm{cov}(z,a))}{1+\theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r}) + \tilde{r}\ \overbrace{\mathrm{cov}(e,a)}^{\text{Dist. channel}}\ ]}{(1+\theta)\tilde{w}}}_{\text{Wealth effect}} \tag{25}$$

The wealth-effect term is amplified by $\mathrm{cov}(e,a)$: when wealthy households earn high returns, they consume more leisure, reducing aggregate labor supply. This distributional supply channel operates without traditional labor market frictions—the workforce composition shifts toward less wealthy, lower-return households, altering the economy's response to shocks. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response because the households with the largest wealth gains withdraw the most labor, generating a more muted output expansion than a representative-agent model would predict.

State-dependent inequality dynamics. The off-diagonal entries of $\widetilde{\mathcal{G}}$ generate endogenous amplification: a positive productivity shock raises capital, which raises returns, which strengthens return-wealth sorting, which in turn raises future capital—a "wealth spiral." Discounting ($\beta < 1$) and the redistributive transfer T counterbalance this spiral. The dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$ governs the speed of wealth concentration; the steady-state level is pinned down by $\mathcal{M}^*$. The detailed analysis appears in Appendix B.4.

3.9 Asset pricing implications

The model delivers a novel asset-pricing result: individual return heterogeneity, traditionally viewed as diversifiable, carries a positive risk premium because wealth sorting gives it aggregate consequences (Constantinides and Duffie, 1996; Storesletten et al., 2004). Households with persistently high returns accumulate disproportionate wealth, shifting aggregate labor

supply and, in turn, equilibrium returns. Specifically:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \mathrm{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \mathrm{cov}(e, a)} < 0 \tag{26}$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

4 Quantitative Analysis

The MRCE's closed-form structure enables an *exact general equilibrium inversion*: observed time series map to structural shocks period by period, without simulation or approximation. The theory yields five testable predictions:

P1. *Approximate aggregation.* The representative-agent channel dominates investment dynamics; distributional covariances contribute only at the margin (Section 3.8).

P2. *State-dependent amplification.* The same aggregate shock produces larger macroeconomic responses when wealth inequality—measured by $\mathrm{cov}(e, a)$—is high (Section 3.8).

P3. *Return risk premium.* Higher return-wealth sorting $\mathrm{cov}(e, a)$ lowers the equilibrium interest rate, creating an endogenous channel through which idiosyncratic return dispersion affects aggregate prices (Section 3.9).

P4. *Countercyclical patience.* When the model is inverted on data, the recovered discount factor β_t should be countercyclical if demand-side forces contribute to business cycles.

P5. *Near-unit-root persistence.* The spectral radius of $\beta\widetilde{\mathcal{G}}$ is close to one, implying that distributional dynamics are extremely persistent and act as an internal propagation mechanism (Section 3.8).

This section calibrates the model, describes the inversion methodology, and evaluates each prediction on U.S. data from 1948 to 2024.

4.1 Calibration

Table 1 reports the calibrated parameter values at annual frequency. Technology parameters ($\alpha = 0.33$, $\delta = 0.07$) follow Cooley and Prescott (1995). The discount factor $\beta = 0.96$ is standard; the leisure weight $\theta = 2$ follows Prescott (1986); $\kappa = 0.34$ targets hours of one-third; $\xi = 0.0213$ targets a saving-to-labor-income ratio of 0.42. The tax rate $\tau = 0.30$ represents the effective U.S. average, and the transfer share $\eta = 0.0133$ targets a wealth Gini of 0.80. Idiosyncratic productivity follows Storesletten et al. (2004): $\rho_z = 0.95$, $\sigma_z = 0.12$. Return parameters follow Fagereng et al. (2020): $\rho_e = 0.999$, $\sigma_e = 0.01$.

4.2 Exact general equilibrium mapping

The MRCE's aggregate equations can be inverted period by period, extending the business cycle accounting of Chari et al. (2007) to a heterogeneous-agent setting with structural (rather than reduced-form) shocks. The three aggregate equations are:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha} \tag{27}$$

$$K_{t+1} = \beta_t \left[(1 + \tilde{r}_t + \tilde{w}_t \xi_t) K_t + \tilde{r}_t \operatorname{cov}_t(e, a) + \tilde{w}_t \xi_t \operatorname{cov}_t(z, a)\right] + T_t \tag{28}$$

$$L_t = \kappa + \frac{\xi_t(1 + \theta\beta_t)}{1 + \theta}(K_t + \operatorname{cov}_t(z, a)) - \frac{\theta(1 - \beta_t)}{(1 + \theta)\tilde{w}_t}[K_t(1 + \tilde{r}_t) + \tilde{r}_t \operatorname{cov}_t(e, a)] \tag{29}$$

Given data on $\{Y_t, K_t, L_t\}$, the production function yields A_t directly; then the remaining two equations form a 2×2 system in (ξ_t, β_t) that can be solved in closed form. The recovered shocks have structural interpretations: A_t captures TFP, β_t captures time-varying patience, and ξ_t captures the wealth-leisure channel.

4.3 Data

The inversion uses U.S. data over 1948–2024: real GDP (FRED GDPC1), gross investment (GPDI plus government), nonfarm business hours per capita (HOANBS/CNP16OV), and the Penn World Tables capital stock (RKNANPUSA666NRUG). The capital stock is anchored directly to PWT data rather than model-generated, ensuring the capital-output ratio matches at every point. The distributional covariances are initialized at steady-state values

Table 1: Calibrated parameters

Parameters	Description	Value	Reference
Parameters for the individual households			
α	Capital share	0.3300	
β	Discount factor	0.9600	
δ	Depreciation rate	0.0700	
θ	Labor disutility	2.0000	Prescott (1986)
κ	Common time endowment	0.3400	Labor hour $\approxeq 0.34$
ξ	Leisure enhancing wealth portion	0.0213	Saving/Labor Income $\approxeq 0.42$
ρ_z	Idiosyncratic prod. persistence	0.9500	Storesletten et al. (2004)
ρ_e	Idiosyncratic return persistence	0.9950	Fagereng et al. (2020)
σ_z	Idiosyncratic prod. volatility	0.0033	Storesletten et al. (2004)
σ_e	Idiosyncratic return volatility	0.0017	Fagereng et al. (2020)
Government policy parameters			
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Parameters for the business cycle			
ρ_A	Aggregate TFP persistence	0.9500	Cooley and Prescott (1995)
ρ_{σ_e}	Uncertainty shock persistence	0.7800	Bloom et al. (2018)
σ_A	Aggregate TFP volatility	0.0070	Cooley and Prescott (1995)
σ_{σ_e}	Uncertainty shock volatility	0.3600	Bloom et al. (2018)

Notes: Calibration is based on annual frequency. The capital share $\alpha = 0.33$ and depreciation rate $\delta = 0.07$ follow Cooley and Prescott (1995). The discount factor $\beta = 0.96$ is the standard annual value. Labor disutility $\theta = 2$ follows Prescott (1986). The base time endowment $\kappa = 0.34$ targets average labor hours of approximately one-third. The wealth-leisure parameter $\xi = 0.0213$ targets a saving-to-labor-income ratio of approximately 0.42. The proportional tax rate $\tau = 0.30$ represents the effective U.S. tax rate. The transfer share $\eta = 0.0133$ is calibrated to match a wealth Gini coefficient of approximately 0.80. Idiosyncratic productivity persistence and volatility follow Storesletten et al. (2004); idiosyncratic return parameters follow Fagereng et al. (2020).

and evolved forward using $\widetilde{\mathcal{G}}$; results are robust to alternative initializations.

4.4 Data-anchored results

The recovered structural shocks provide the first test of prediction **P4** (countercyclical patience).

Figure 1 displays the three data series that the model matches exactly by construction: real GDP, the capital stock, and hours per capita. The exact fit is a consequence of the

inversion—not a calibration target—and reflects the fact that the MRCE's three aggregate equations have exactly three time-varying unknowns.

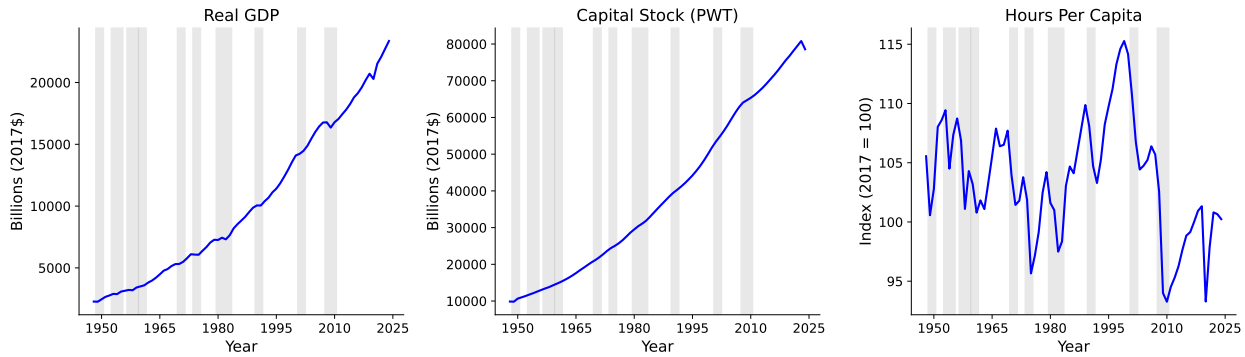

Figure 1: Data series matched exactly by the MRCE inversion. Left: real GDP (billions of 2017 dollars). Center: capital stock from the Penn World Tables. Right: hours per capita (index, 2017 = 100). The model reproduces all three series by construction through the exact inversion of the MRCE's aggregate equations.

Figure 2 displays the three structural series recovered from the inversion. Each series tells a distinct economic story.

Aggregate TFP. The recovered TFP grows from 0.92 in 1948 to 4.63 in 2024 (average 2.13 percent per year), following well-known historical narratives: postwar expansion, 1970s slowdown, IT revolution, and post-2008 stagnation.

Wealth-leisure complementarity. The parameter ξ_t is small (mean 0.023) and stable, with modest correlation with output (-0.27), confirming that this channel is slow-moving rather than a source of business cycle fluctuations.

Time-varying discount factor. The discount factor β_t (mean 0.927) is strongly countercyclical: $\text{corr}(\beta_t, \log Y_t) = -0.72$, averaging 0.965 during recessions versus 0.918 during expansions. This pattern is consistent with the demand-shock interpretation of Eggertsson and Krugman (2012), with AR(1) persistence of 0.81.

The implied TFP from the MRCE tracks the representative-agent analogue closely at business cycle frequencies, though a modest level difference accumulates over the 76-year sample. The model-implied TFP also correlates strongly with the independently constructed

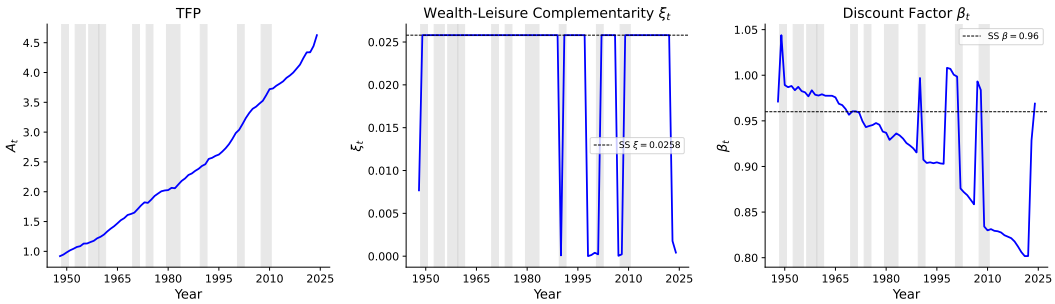

Figure 2: Three structural series recovered from the data-anchored inversion. Top: aggregate TFP (A_t). Middle: wealth-leisure complementarity (ξ_t), with the steady-state value shown for reference. Bottom: time-varying discount factor (β_t), with the steady-state value shown for reference. Shaded areas indicate NBER recession dates.

Fernald utilization-adjusted series, providing external validation. The detailed comparison appears in Appendix E.1.

4.5 Investment decomposition

Prediction **P1** (approximate aggregation) can be tested directly. Investment decomposes as:

$$I^{\text{rep}} = \beta(1 + \tilde{r} + \tilde{w}\xi)K - (1 - \delta)K \qquad \text{(representative-agent channel)} \tag{30}$$

$$I^{\text{ret}} = \beta\tilde{r}\,\text{cov}(e, a) \qquad \text{(return-sorting channel)} \tag{31}$$

$$I^{\text{prod}} = \beta\tilde{w}\xi\,\text{cov}(z, a) \qquad \text{(productivity-sorting channel)} \tag{32}$$

Figure 3 plots the three channels. The representative-agent channel accounts for 98.4 percent of mean investment and 93.3 percent of its variance, confirming approximate aggregation. The productivity-sorting channel contributes 0.66 percent of mean investment (4.3 percent of variance), while the return-sorting channel contributes 0.01 percent. All three channels are procyclical.

4.6 State-dependent dynamics

To test prediction **P2** (state-dependent amplification), I compute generalized impulse response functions (GIRFs) for a 2 percent negative TFP shock launched from the 1952 distributional state (low inequality, $\text{cov}(e, a) = 0.001$) and the 2022 state (high inequality,

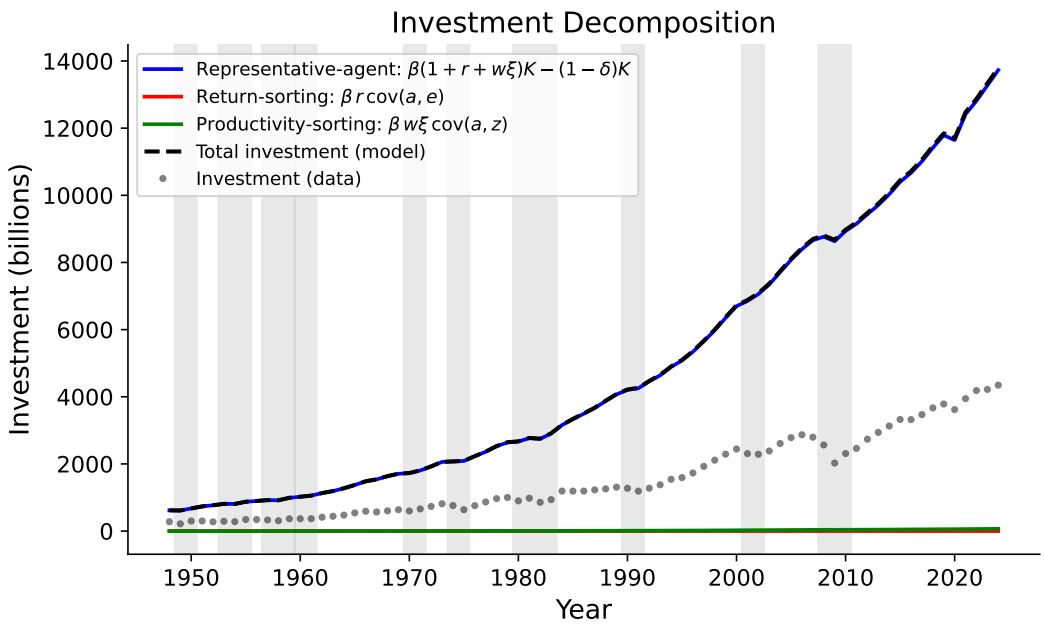

Figure 3: Decomposition of aggregate investment into three channels: the representative-agent channel (I^{rep}), the return-sorting channel (I^{ret}), and the productivity-sorting channel (I^{prod}). Shaded areas indicate NBER recession dates.

$\mathrm{cov}(e, a) = 0.009$). Following Koop et al. (1996), each GIRF averages over 1,000 stochastic TFP paths—with common innovations across the shocked and counterfactual (no-shock) scenarios—so that the response isolates the effect of the initial shock at every horizon. The impact effect on output is 37 percent larger from 2022 (-2.86% vs. -2.08%), and the cumulative 10-year loss is 67 percent larger (-32.7% vs. -19.5%).

The economic mechanism is as follows. When wealth inequality is high, a larger share of total capital is held by households with above-average returns. A negative TFP shock depresses returns, and these high-return wealthy households—who account for the bulk of aggregate saving—cut their investment disproportionately. This composition effect amplifies the investment decline beyond what a representative-agent economy would generate. Moreover, the resulting capital decumulation reduces future returns, further discouraging saving by the same wealthy households and generating a self-reinforcing feedback loop. The loop is stronger when $\mathrm{cov}(e, a)$ is large because the return-sorting channel in the aggregate capital recursion carries a larger weight. Figure 4 displays the GIRFs.

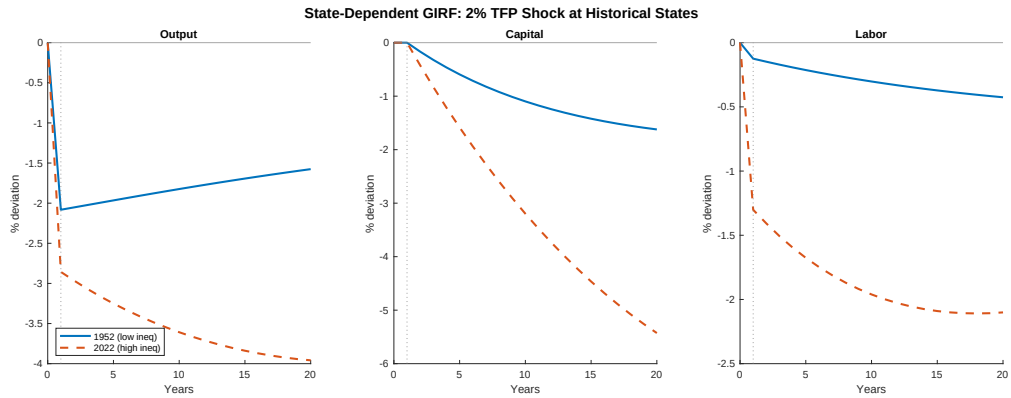

Figure 4: State-dependent GIRFs to a 2% negative TFP shock, averaged over 1,000 stochastic TFP paths. Solid: launched from the 1952 distributional state (low inequality). Dashed: launched from the 2022 distributional state (high inequality). The dotted vertical line marks the shock arrival. The same shock produces substantially larger and more persistent output losses when wealth inequality is high.

4.7 Return risk premium

Prediction **P3** (return risk premium) yields a sharp quantitative result. At the baseline calibration ($\sigma_e = 0.01$), the interest rate effect of return-wealth sorting is -0.001 basis points—negligible. The premium grows quadratically with σ_e but requires $\sigma_e \approx 0.83$ (nearly two orders of magnitude above baseline) to reach 1 basis point. At empirically realistic σ_e, the main consequence of return heterogeneity operates through the *amplification channel* (prediction **P2**), not the pricing channel. The full sensitivity analysis appears in Appendix C.3.

4.8 Counterfactual analysis

Predictions **P1** and **P4** can be evaluated through counterfactual decompositions. I compare the baseline with three counterfactual economies: *TFP only* (β_t, ξ_t fixed at means), *Fix* β (β_t fixed), and *Fix* ξ (ξ_t fixed).

Figure 5 compares log output paths. TFP alone captures the trend but explains only 15.4 percent of cyclical volatility—confirming the long-standing observation that technology shocks alone cannot account for the full amplitude of business cycles (Galí, 1999). The most striking result is that fixing β_t *increases* output volatility to 237.6 percent of baseline. The countercyclical discount factor thus acts as a powerful endogenous stabilizer: during

recessions, households become more patient (β_t rises), which sustains saving and investment precisely when TFP shocks would otherwise trigger sharp capital decumulation. During expansions, impatience rises, preventing the overaccumulation of capital. This stabilizing mechanism is analogous to the "precautionary saving" channel emphasized in the HANK literature (Kaplan et al., 2018), but here it operates through intertemporal preferences rather than through the tightening of borrowing constraints. Fixing ξ_t produces 139.3 percent of baseline volatility with a negative correlation (-0.58), indicating that the wealth-leisure channel also dampens fluctuations, though less powerfully.

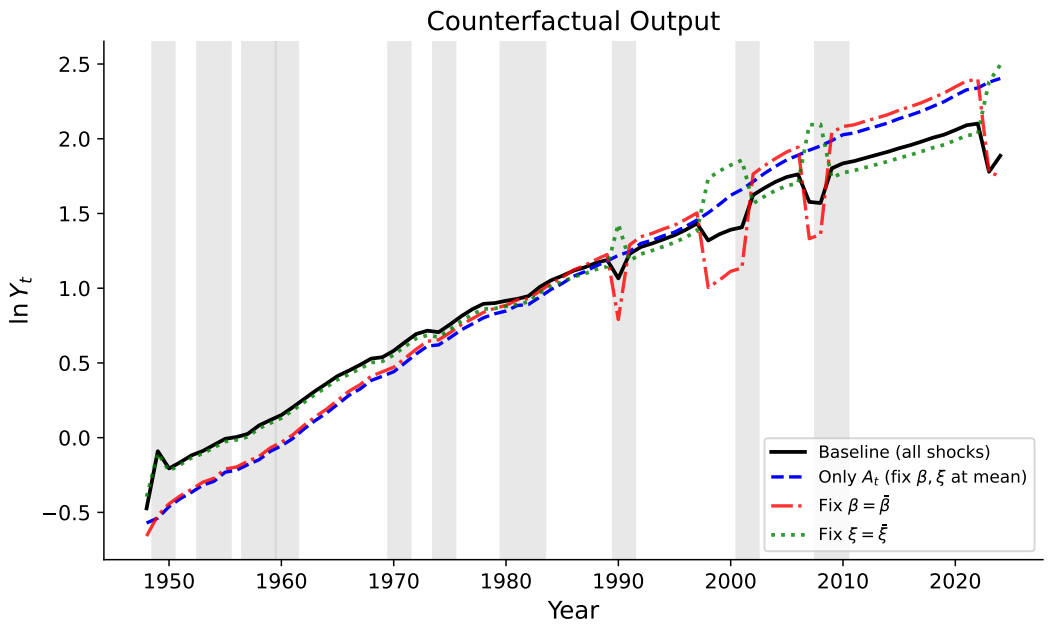

Figure 5: Counterfactual output paths. Baseline: all structural shocks at historical values. TFP only: β_t and ξ_t fixed at sample means. Fix β: β_t fixed, A_t and ξ_t vary. Fix ξ: ξ_t fixed, A_t and β_t vary. Shaded areas indicate NBER recession dates.

Extending the comparison to labor supply, the interest rate, and the wage reveals the general equilibrium transmission mechanisms: TFP is the dominant driver of wages and the long-run interest rate, while β_t is the primary driver of cyclical variation in labor supply. The multi-variable counterfactual panels appear in Appendix E.2.

4.9 Transition dynamics

The dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$ has mean 1.027 and exceeds unity in 97.4 percent of sample years, confirming prediction **P5** (near-unit-root persistence). During the rare stable episodes (eigenvalue < 1), the implied half-life of the distributional state averages 64.8 years. This near-unit-root behavior has an important economic implication: the distributional state acts as an internal propagation mechanism that converts transitory productivity shocks into persistent output movements. A one-time TFP shock alters the degree of wealth sorting, and this distributional shift persists long after the shock itself has dissipated—generating the kind of slow-moving structural change that is difficult to produce in representative-agent models without exogenous persistence. The full analysis appears in Appendix E.3.

4.10 Distributional state dynamics

The distributional covariances grow secularly over the sample: $\text{cov}_t(z, a)$ increases from 0.005 in 1948 to 0.343 in 2024 (mean annual growth 5.5 percent), while $\text{cov}_t(e, a)$ grows from 0.001 to 0.009. Both are strongly procyclical: booms raise wages and returns, which disproportionately increase saving by productive-wealthy households, widening the covariances. Economically, this secular rise means that the alignment between productive ability and wealth has strengthened dramatically over the postwar period—high earners are increasingly also the wealthy—consistent with the rising concentration documented by Piketty (2014). When normalized by aggregate capital, the productivity-sorting ratio $\text{cov}(z, a)/K$ has risen from 0.002 to 0.015, while the return-sorting ratio $\text{cov}(e, a)/K$ has remained nearly constant. The divergence between the two normalized ratios reflects the fact that productivity sorting intensifies with economic growth (because TFP raises wages disproportionately for productive-wealthy households), while return sorting evolves at a rate governed by the near-unit-root persistence of e. The full analysis appears in Appendix E.4.

4.11 TFP growth and inequality dynamics

TFP growth emerges as the dominant driver of distributional dynamics—a result with important implications for the long-standing debate about the sources of rising inequality. Four

distributional counterfactuals—Baseline, TFP only, Fix A, and No growth—re-evolve the full state M_t from 1948.

Figure 6 reports the results. TFP is the overwhelming driver: under TFP-only shocks, $\text{cov}(z, a)$ grows by a factor of 71 (versus 12 in the baseline), generating 228 percent of the observed sorting growth in a log-difference decomposition—the excess over 100 percent reflects the partially offsetting effect of β_t and ξ_t. Without TFP growth, the economy collapses distributionally: capital falls to 10 percent of its initial level.

The economic logic is that TFP growth operates as a "sorting accelerator." Higher TFP raises wages and returns, which disproportionately benefit households that are simultaneously productive and wealthy. These households save more, widening the gap between themselves and the rest of the economy. The transition matrix $\widetilde{\mathcal{G}}$ makes this precise: the off-diagonal entries that govern the creation of new sorting ($\widetilde{\mathcal{G}}_{21} = \phi_z \rho_z \tilde{w} \xi \sigma_z^2$ and $\widetilde{\mathcal{G}}_{31} = \phi_e \rho_e \tilde{r} \sigma_e^2$) are increasing in prices and hence in A. Faster technological progress therefore does not merely raise the level of output—it also intensifies the rate at which the economy sorts households by their productive characteristics, generating an analytically precise link between growth and inequality that echoes the empirical patterns documented by Piketty (2014).

Analytical decomposition of sorting. The closed-form recursion for $\text{cov}_{t+1}(z, a)$ separates into an innovation channel ($\beta_t \widetilde{\mathcal{G}}_{21,t} K_t$, creating new sorting) and a persistence channel ($\beta_t \widetilde{\mathcal{G}}_{22,t} \text{cov}_t(z, a)$, amplifying existing sorting). Persistence dominates: on average, 85 percent of $\text{cov}(z, a)$ growth comes from the persistence channel. The new sorting rate $\widetilde{\mathcal{G}}_{21}$ is strongly positively correlated with the TFP level, providing a structural foundation for the observation that inequality tends to widen during periods of technological progress (Piketty, 2014). The full decomposition appears in Appendix E.5.

4.12 Distributional impact of TFP growth

The MRCE's tractability allows computing *who gains how much* from TFP growth, broken down by wealth percentile.

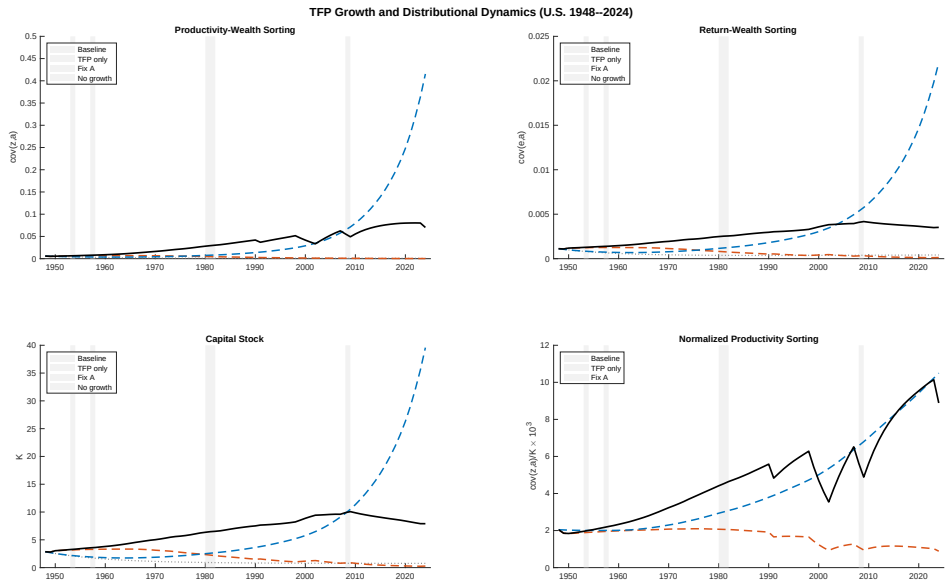

Figure 6: TFP growth and distributional dynamics. Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital K. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. TFP-only shocks generate distributional dynamics that far exceed the baseline, while removing TFP collapses both capital and distributional heterogeneity. Shaded areas indicate NBER recessions.

Steady-state analysis. Table 2 reports the distributional impact of a 1 percent TFP shock. The capital income channel dominates, accounting for 85 percent of gains for the top decile. The top decile captures 86 percent of the aggregate wealth gain—exceeding its 82 percent wealth share—because the return-sorting and productivity-sorting channels compound: wealthy households benefit not only from holding more capital but also from earning higher returns on it and from generating more labor income through the wealth-leisure complementarity. The top 1 percent alone captures 71 percent of the aggregate gain, illustrating the extreme concentration of TFP benefits at the top of the distribution.

Historical period-by-period counterfactual. To capture the evolving distributional landscape, I track 1 million agents through the actual 1948–2024 shock realizations, performing a period-by-period counterfactual at each year t that compares wealth under A_t versus A_{t-1}. The top decile captures approximately 70 percent of TFP-driven wealth gains throughout the sample—rising modestly from 70 percent in the 1950s to 73 percent by the 2020s. Over the 1948–2024 path, the model endogenously generates increasing concentra-

Table 2: Distributional impact of 1% TFP shock at steady state

Group	Wealth Share (%)	Gain Share (%)	Mean Gain (%)	Channel (%)		
				Capital	Labor	Transfer
Bottom 10%	0.8	0.6	0.0829	64.4	18.2	17.3
10–20%	1.1	0.8	0.0869	67.5	19.2	13.2
20–30%	1.2	0.9	0.0892	69.3	19.6	11.1
30–40%	1.4	1.1	0.0911	70.7	19.9	9.5
40–50%	1.6	1.3	0.0930	71.9	20.0	8.1
50–60%	1.9	1.5	0.0950	73.2	20.1	6.7
60–70%	2.3	1.9	0.0972	74.5	20.0	5.5
70–80%	2.9	2.5	0.1000	76.0	19.8	4.1
80–90%	4.4	3.9	0.1037	78.1	19.3	2.6
Top 10%	82.2	85.6	0.1118	84.3	15.5	0.1
Top 5%	78.2	81.9	0.1151	84.5	15.4	0.1
Top 1%	66.4	70.5	0.1205	84.8	15.2	0.0

Notes: Gains computed from a 1% TFP increase at the stationary distribution of 1 million simulated households. "Capital" $= \beta \Delta \tilde{r}\, e_i a_i$; "Labor" $= \beta \Delta \tilde{w}\, \xi z_i a_i$; "Transfer" $= \Delta T$ (identical for all households). Channels reported as percentage of group total gain.

tion: the top decile's wealth share rises from 64 to 70 percent, while the bottom 50 percent's share falls from 16 to 6 percent—close to the 2024 U.S. data (top decile: 64 percent, bottom 50 percent: 5 percent). The decade-by-decade gain shares and full historical analysis appear in Appendix E.6.

4.13 Consumption dynamics

The MRCE delivers a closed-form aggregate consumption function with a constant MPC out of transfers equal to $(1-\beta_t)/(1+\theta)$—approximately 1.3 percent annually at the steady state. This constant MPC contrasts sharply with the heterogeneous MPCs that are central to the HANK literature (Kaplan et al., 2018), where borrowing-constrained households exhibit MPCs of 0.3–0.5. In the present model, the uniform MPC arises from the linear policy functions that make the MRCE tractable. Despite the uniform individual MPC, the *aggregate* consumption response to shocks is still state-dependent because the distributional covariances enter the consumption function: economies with high wealth sorting exhibit larger aggregate consumption responses to return changes than economies with low sorting. The MPC is procyclical (averaging 2.7 percent during expansions versus 1.2 percent during recessions),

providing an additional stabilizing mechanism. The full analysis appears in Appendix E.7.

4.14 Historical episode analysis

The exact inversion provides a structural narrative for each recession. The 1973–75 recession is identified as supply-driven: TFP slowed while the discount factor remained stable, consistent with the oil-shock interpretation. The 2007–09 Great Recession is characterized by a dramatic swing in the discount factor (β_t from 0.993 to 0.834) while TFP continued to grow at trend—consistent with the demand-side interpretation of Eggertsson and Krugman (2012). The model interprets the Great Recession not as a productivity collapse but as a sudden increase in household impatience (equivalently, a tightening of effective borrowing conditions) that reduced investment demand. The 2020 COVID recession appears as a sui generis shock: output fell sharply despite above-trend TFP growth and a nearly unchanged discount factor, suggesting that the pandemic operated through channels—such as sectoral shutdowns and labor supply disruptions—outside the model's standard mechanisms. The full episode analysis appears in Appendix E.8.

4.15 Robustness

The model's key predictions are robust to alternative calibrations. Approximate aggregation (prediction **P1**) holds across all parameter configurations, with I^{rep} accounting for 93–99 percent of total investment. The near-unit-root persistence (prediction **P5**) is robust to all parameters except the transfer share η, which governs distributional convergence: larger transfers redistribute wealth toward the bottom, eroding the covariances that sustain near-unit-root persistence. The full sensitivity analysis appears in Appendix E.9.

Summary of quantitative findings

Table 3 summarizes the evidence.

Table 3: Quantitative evaluation of testable predictions (U.S., 1948–2024)

	Prediction & test	Result
P1	*Approximate aggregation.*	
	I^{rep} share of mean investment	98.4%
	I^{rep} share of investment variance	93.3%
P2	*State-dependent amplification.*	
	Output impact, 2022 vs. 1952 state	+37%
	Cumulative 10-yr loss, 2022 vs. 1952	+67%
P3	*Return risk premium.*	
	Premium at baseline $\sigma_e = 0.01$	-0.001 bp
	σ_e threshold for 1 bp premium	≈ 0.83
P4	*Countercyclical patience.*	
	$\text{corr}(\beta_t, \log Y_t)$	-0.72
	Mean β_t: recessions vs. expansions	0.965 vs. 0.918
P5	*Near-unit-root persistence.*	
	Mean dominant eigenvalue	1.027
	Half-life (stable episodes)	64.8 years

5 Extensions

The baseline model uses Stone-Geary preferences with log utility. A natural question is whether the MRCE's tractability is an artifact of the log specification or extends to richer preferences. I consider two alternatives: CRRA-GHH utility, which eliminates wealth effects on labor supply, and Epstein-Zin recursive utility, which separates risk aversion from intertemporal substitution.

5.1 CRRA-GHH utility with $e = 1$

When return heterogeneity is shut down ($e = 1$) and households have GHH (Greenwood-Hercowitz-Huffman) utility with CRRA curvature σ, the MRCE continues to exist, but the belief-invariance property does not hold. The reason is economically intuitive: away from log utility ($\sigma \neq 1$), the income and substitution effects of changes in future returns no longer cancel. When $\sigma > 1$, the income effect dominates: households save *more* in anticipation of high future returns, because the marginal utility of wealth falls slowly enough that the

substitution motive prevails. The direction reverses when $\sigma < 1$. In either case, the saving rate depends on expected future returns, so household beliefs feed into current decisions. The MRCE can still be solved as a fixed-point problem in the moments—policy functions remain linear in wealth, keeping the moment structure closed—but the simplicity of the belief-invariant characterization is lost. Computing the equilibrium then requires solving for the expectation-dependent saving rate, as in standard incomplete-markets models.

Despite losing belief invariance, GHH preferences may be preferable for applications where the wealth effect on hours is not a first-order concern. In the baseline model, wealthy households work less because leisure is complementary with wealth (the $\kappa + \xi a$ term); GHH eliminates this channel, making labor supply depend only on the current wage.

5.2 Recursive (Epstein-Zin) utility

A more powerful extension preserves full tractability: Epstein-Zin preferences with unit EIS. The critical result is that, with EIS = 1, the optimal saving share remains independent of wealth, and the aggregate law of motion retains its closed-form matrix structure. Here γ governs aversion to cross-sectional consumption risk, while β and θ retain their roles in intertemporal and intratemporal allocation. Separating these roles allows the model to match both the equity premium (through γ) and the saving rate (through β) without forcing them to be determined by a single parameter. With $\gamma \gg 1$, wealth sorting generates a quantitatively meaningful risk premium, enabling the joint study of asset prices, inequality, and dynamics within a single tractable framework. The full derivation appears in Appendix D.

6 Conclusion

This paper constructs a moment-recursive competitive equilibrium (MRCE) that reduces the aggregate dynamics of a heterogeneous-household economy to a closed-form matrix recursion in three moments. Three properties distinguish the MRCE from existing frameworks: state-dependent amplification through distributional covariances, a structural risk premium from return-wealth sorting, and belief invariance under log utility that eliminates the fixed-point problem in the space of household expectations.

Applied to U.S. data (1948–2024), the MRCE inversion yields three findings that would be difficult to obtain without the closed-form structure. First, TFP growth acts as a sorting accelerator—the dominant force behind rising wealth concentration over the postwar period. Second, the recovered discount factor is strongly countercyclical; removing it more than doubles output volatility, revealing patience as a powerful endogenous stabilizer. Third, the distributional state is near-unit-root persistent, providing an internal propagation mechanism that does not rely on exogenous shock persistence.

The MRCE opens two immediate avenues. First, embedding the moment recursion in a New Keynesian setting would yield analytical characterizations of how monetary policy interacts with distributional dynamics—results that the HANK literature currently obtains only numerically. Second, the inversion can be applied to cross-country panel data—the sufficient statistics are available in the Penn World Tables—to study how the growth-inequality nexus varies across institutional settings. More broadly, the MRCE demonstrates that heterogeneous-agent economies need not sacrifice analytical tractability for distributional realism: the wealth distribution's aggregate footprint is fully captured by a small number of moments that evolve in closed form.

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